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Report History

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Contents

1 Introduction

This lab was all about building a smart room controller using digital logic. The goal was to manage lighting and blinds based on four different scenes: Standard, Lunchtime, Nighttime, and Emergency. We started by designing the logic and testing it through simulations, then moved on to writing VHDL code and running it on real hardware using an FPGA and IOM board.

2 Summary of Preparatory Tasks

2.1 Preparation Task 1: Logic Derivation

The first preparatory step was to determine the logic of the Smart Home Room Controller system using five inputs: `LSW` (Lamp Switch), `BSW` (Blind Switch), `SENS` (Sunlight Sensor), and two scene selector inputs: `S0` and `S2`. The outputs are `Lamp` and `Blind`.

The following Boolean expressions were derived:

$$\begin{aligned}
\text{Lamp} &= (\overline{\text{BSW}} \land \text{LSW} \land \overline{\text{S2}}) \lor (\text{LSW} \land \overline{\text{S2}} \land \overline{\text{SENS}}) \lor (\text{LSW} \land \text{S0}) \lor (\text{S0} \land \text{S2}) \\
\text{Blind} &= (\text{BSW} \land \overline{\text{S0}} \land \overline{\text{S2}}) \lor (\text{S0} \land \text{S2}) \lor (\text{S2} \land \overline{\text{SENS}})
\end{aligned}$$

The Boolean behavior was analyzed using a comprehensive truth table.

LSW	BSW	$SENS$	$S0$	$S2$	$Lamp$	$Blind$
0	0	0	0	0	0	0
0	0	0	0	1	0	1
0	0	0	1	0	0	0
0	0	0	1	1	1	1
0	0	1	0	0	0	0
0	0	1	0	1	0	0
0	0	1	1	0	0	0
0	0	1	1	1	1	1
0	1	0	0	0	0	1
0	1	0	0	1	0	1
0	1	0	1	0	0	0
0	1	0	1	1	1	1
0	1	1	0	0	0	1
0	1	1	0	1	0	0
0	1	1	1	0	0	0
0	1	1	1	1	1	1
1	0	0	0	0	1	0
1	0	0	0	1	0	1
1	0	0	1	0	1	0
1	0	0	1	1	1	1
1	0	1	0	0	1	0
1	0	1	0	1	0	0
1	0	1	1	0	1	0
1	0	1	1	1	1	1
1	1	0	0	0	1	1
1	1	0	0	1	0	1
1	1	0	1	0	1	0
1	1	0	1	1	1	1
1	1	1	0	0	0	1
1	1	1	0	1	0	0
1	1	1	1	0	1	0
1	1	1	1	1	1	1

2.2 Preparation Task 2: Logic Design in Digital

The logic expressions were implemented in the *Digital* software environment. Two versions of the controller were created:

- A version using basic gates (AND, OR, NOT), as shown in Figure 1
- A version using only NAND gates, shown in Figure 2

Both circuits were tested to verify their correctness through simulation.

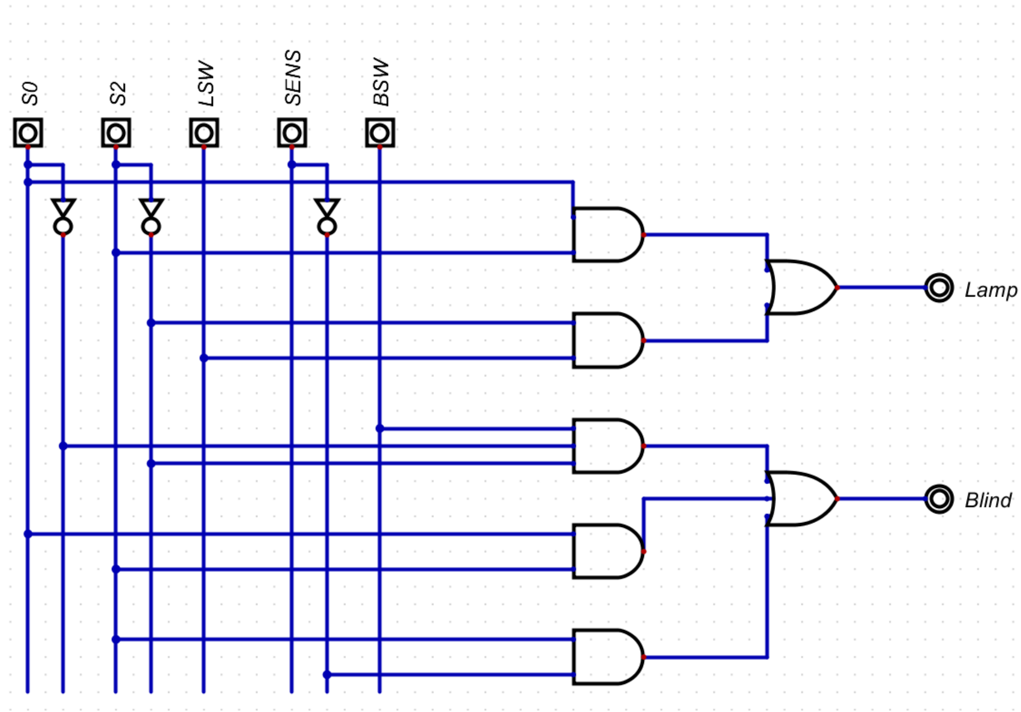

Figure 1: AND/OR/NOT Logic Design of the Room Controller

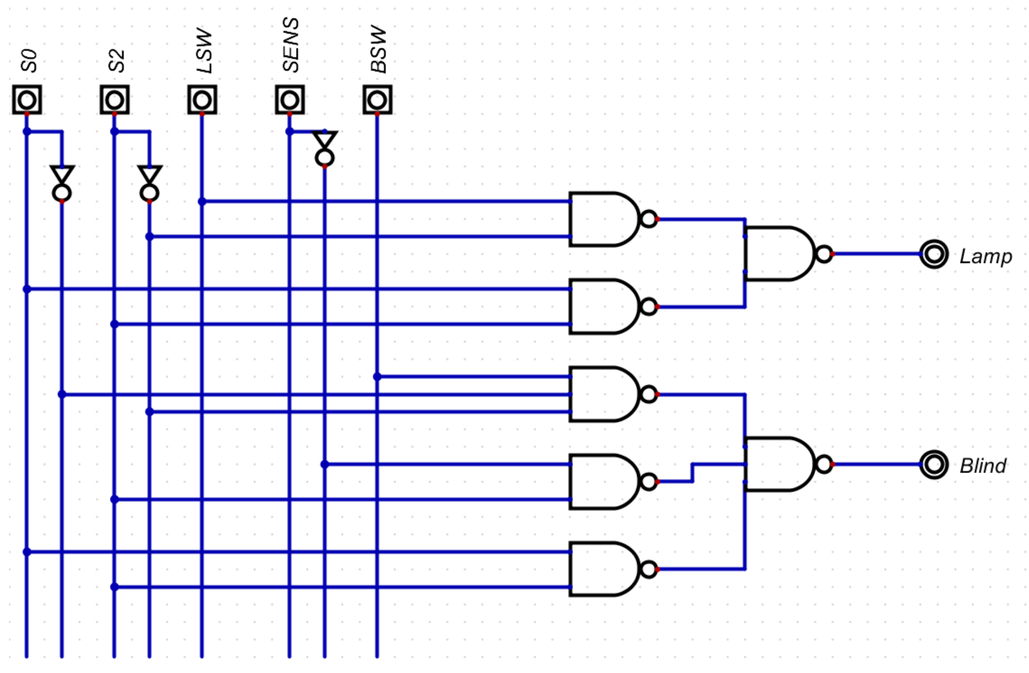

Figure 2: NAND-only Logic Design of the Room Controller

2.3 Preparation Task 3: VHDL Simulation and Test Plan

The Boolean logic was translated into VHDL code. The following code shows the final architecture used in simulation:

```
architecture Behavioral of main is
  signal s1: std_logic;
begin
  s1 <= NOT S2;
  Lamp <= NOT (NOT (LSW AND s1) AND NOT (S0 AND S2));
  Blind <= NOT (NOT (BSW AND NOT S0 AND s1) AND NOT (NOT SENS AND S2)
  AND NOT (S0 AND S2));
end Behavioral;
```

A comprehensive testbench was written to evaluate all 16 major scenarios across all scenes. Each test is structured to match Scene[1:0] combinations and control input permutations:

- Scene 00 → R1.1 to R1.4
- Scene 01 → R2.1 to R2.4
- Scene 10 → R3.1 to R3.4
- Scene 11 → R4.1 to R4.4

Simulation was performed in both *Digital* and EDAPlayground. Waveform outputs confirmed that the Lamp and Blind signals matched expectations for every input combination.

2.4 Testbench Excerpt

```
-- Scene = 00 (S2=0, S0=0)
R1.1: LSW <= '0'; BSW <= '0'; SENS <= '0'; S0 <= '0'; S2 <= '0'; wait for 10 ns;
R1.2: LSW <= '1'; BSW <= '0'; SENS <= '1'; S0 <= '0'; S2 <= '0'; wait for 10 ns;
R1.3: LSW <= '0'; BSW <= '1'; SENS <= '1'; S0 <= '0'; S2 <= '0'; wait for 10 ns;
R1.4: LSW <= '1'; BSW <= '1'; SENS <= '0'; S0 <= '0'; S2 <= '0'; wait for 10 ns;

-- Scene = 01 (S2=0, S0=1)
R2.1: LSW <= '1'; BSW <= '0'; SENS <= '1'; S0 <= '1'; S2 <= '0'; wait for 10 ns;
R2.2: LSW <= '0'; BSW <= '0'; SENS <= '0'; S0 <= '1'; S2 <= '0'; wait for 10 ns;
R2.3: LSW <= '1'; BSW <= '1'; SENS <= '0'; S0 <= '1'; S2 <= '0'; wait for 10 ns;
R2.4: LSW <= '0'; BSW <= '1'; SENS <= '1'; S0 <= '1'; S2 <= '0'; wait for 10 ns;

-- Scene = 10 (S2=1, S0=0)
R3.1: LSW <= '1'; BSW <= '0'; SENS <= '1'; S0 <= '0'; S2 <= '1'; wait for 10 ns;
R3.2: LSW <= '0'; BSW <= '0'; SENS <= '0'; S0 <= '0'; S2 <= '1'; wait for 10 ns;
R3.3: LSW <= '1'; BSW <= '1'; SENS <= '0'; S0 <= '0'; S2 <= '1'; wait for 10 ns;
R3.4: LSW <= '0'; BSW <= '1'; SENS <= '1'; S0 <= '0'; S2 <= '1'; wait for 10 ns;

-- Scene = 11 (S2=1, S0=1)
R4.1: LSW <= '0'; BSW <= '0'; SENS <= '0'; S0 <= '1'; S2 <= '1'; wait for 10 ns;
R4.2: LSW <= '1'; BSW <= '1'; SENS <= '0'; S0 <= '1'; S2 <= '1'; wait for 10 ns;
R4.3: LSW <= '0'; BSW <= '1'; SENS <= '1'; S0 <= '1'; S2 <= '1'; wait for 10 ns;
R4.4: LSW <= '1'; BSW <= '0'; SENS <= '1'; S0 <= '1'; S2 <= '1'; wait for 10 ns;
```

2.5 Test Plan

Req. ID	Test description	stimuli	expected response	"Digital" Simulation (check)	coded in Testbench (line number range)	Testbench Simulation (check)	7400 circuit (check)	FPGA circuit (check)
R1.1	Standard Scene Case 1	So=0,S2=0, SE=0 BSW=0,LSW=0	L=0,B=0	✓	40-41	✓	✓	✓
R1.2	Standard Scene Case 2	So=0,S2=0, SE=1 BSW=0,LSW=1	L=1,B=0	✓	42-43	✓	✓	✓
R1.3	Standard Scene Case 3	So=0,S2=0, SE=1 ,BSW=1,LSW=0	L=0,B=1	✓	44-45	✓	✓	✓
R1.4	Standard Scene Case 4	So=0,S2=0, SE=0 BSW=1,LSW=1	L=1,B=1	✓	46-47	✓	✓	✓
R2.1	Lunchtime Scene Case 1	So=0,S2=1, SE=1 ,BSW=0,LSW=1	L=0,B=0	✓	49-50	✓	✓	✓
R2.2	Lunchtime Scene Case 2	So=0,S2=1, SE=0 ,BSW=0,LSW=0	L=0,B=1	✓	51-52	✓	✓	✓
R2.3	Lunchtime Scene Case 3	So=0,S2=1, SE=0 ,BSW=1,LSW=1	L=0,B=1	✓	53-54	✓	✓	✓
R2.4	Lunchtime Scene Case 4	So=0,S2=1, SE=1 ,BSW=1,LSW=0	L=0,B=0	✓	55-56	✓	✓	✓
R3.1	Nighttime Scene Case 1	So=1,S2=0, SE=1 ,BSW=0,LSW=1	L=1,B=0	✓	58-59	✓	✓	✓
R3.2	Nighttime Scene Case 2	So=1,S2=0, SE=0 ,BSW=0,LSW=0	L=0,B=0	✓	60-61	✓	✓	✓
R3.3	Nighttime Scene Case 3	So=1,S2=0, SE=0 ,BSW=1,LSW=1	L=1,B=0	✓	62-63	✓	✓	✓
R3.4	Nighttime Scene Case 4	So=1,S2=0, SE=1 ,BSW=1,LSW=0	L=0,B=0	✓	64-65	✓	✓	✓
R4.1	Emergency Scene Case 1	So=1,S2=1, SE=0 ,BSW=0,LSW=0	L=1,B=1	✓	67-68	✓	✓	✓
R4.2	Emergency Scene Case 2	So=1,S2=1, SE=0 ,BSW=1,LSW=1	L=1,B=1	✓	69-70	✓	✓	✓
R4.3	Emergency Scene Case 3	So=1,S2=1, SE=1 ,BSW=1,LSW=0	L=1,B=1	✓	71-72	✓	✓	✓
R4.4	Emergency Scene Case 4	So=1,S2=1, SE=1 ,BSW=0,LSW=1	L=1,B=1	✓	73-74	✓	✓	✓

Figure 3: Test cases (R1.1–R4.4 cases)

3 Task 1: Lamp Control Implementation with NAND Gates

In this task, we constructed the "Lamp" portion of the smart room controller using only 2-input NAND gates. The circuit was implemented on a solderless evaluation board using a 7400 quad-NAND IC and was tested under multiple scenarios.

Objective

To replicate the logic controlling the Lamp output, derived from the main truth table, using only NAND gates. The circuit was validated both in simulation and on a physical board using LEDs and switches.

Truth Table and Boolean Expression

The Lamp logic depends on three inputs: `SC1`, `SC2`, and `SWITCH`. The resulting truth table is shown below:

$SC1$	$SC2$	$SWITCH$	$LIGHT$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

$$LIGHT = (SC1 \wedge SC2) \vee (\overline{SC2} \wedge SWITCH)$$

Simulation with Digital Software

We simulated this logic in the "Digital" software using only NAND gates. The circuit was functionally verified against the truth table above.

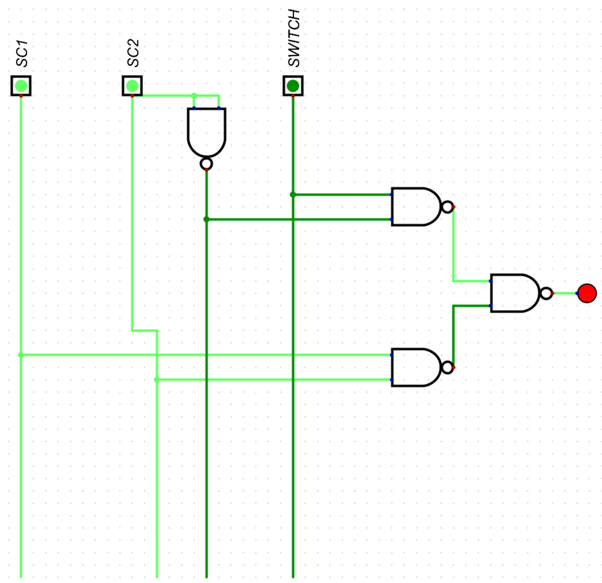

Figure 4: NAND Gate Simulation of Lamp Logic in Digital

7400 IC Layout

The simulation was then translated to an IC-based layout using the 7400 quad-NAND chip. Each gate was connected according to its role in the logic expression. Inputs were driven via digital switches and output was monitored using an LED.

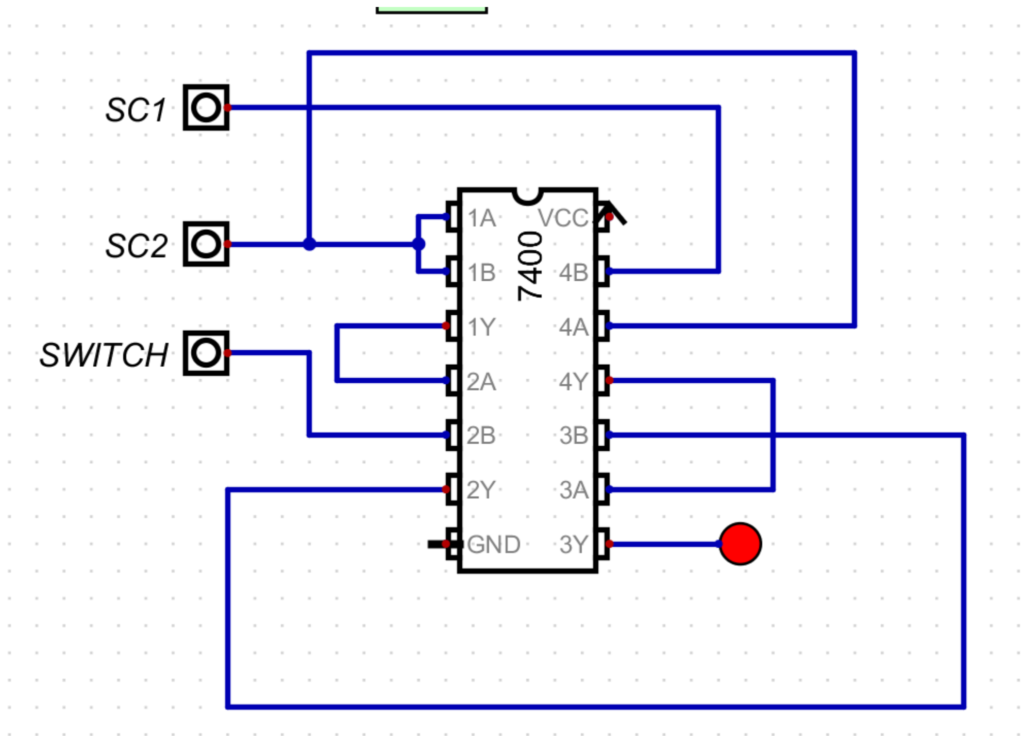

Figure 5: Pin-level Simulation using 7400 NAND IC in Digital

Physical Implementation

The finalized circuit was physically constructed on a TC Evaluation Board. Inputs were wired through toggle switches, and a red LED was used to indicate the Lamp output.

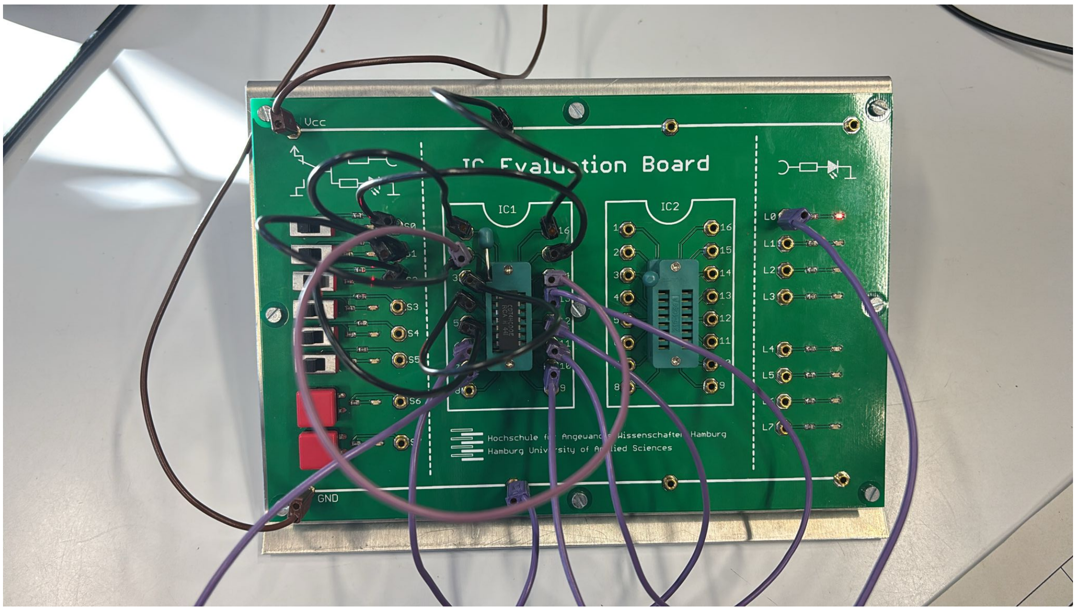

Figure 6: Real-life implementation on evaluation board (LED ON for SC1=1, SC2=1)

The LED turned on exactly as expected during test case SC1=1, SC2=1, SWITCH=0 — confirming that the physical circuit behaves according to the logic design.

Notes on Hardware

- The IC used was a standard 7400 quad 2-input NAND DIP.
- GND (Pin 7) and Vcc (Pin 14) were connected to 0 V and 3.3 V respectively.
- Proper orientation of the chip and socket was double-checked to avoid damage.

Conclusion: The lamp logic function was successfully realized using only NAND gates. Simulations and physical circuit results matched the theoretical truth table for all tested input combinations.

4 Task 2.1: Signal Generator and Oscilloscope Setup

In this task, the setup for dynamic testing of the NAND-based Lamp control circuit was prepared. The focus was to connect a function generator as an input stimulus and configure an oscilloscope to capture the resulting output behavior.

Setup Details

- **Signal Generator:**
  - Waveform: Square wave
  - Amplitude: 3.3 V peak-to-peak
  - Offset: 1.65 V
  - Frequency: 10 kHz
- **Connections:**
  - Signal generator output connected to the Lamp control circuit input (replacing manual switch)
  - Oscilloscope CH1 connected to the signal generator output (input signal)
  - Oscilloscope CH2 connected to the output of the NAND Lamp circuit (LED node)

This configuration enabled the measurement of propagation delay by comparing input and output waveforms on the oscilloscope.

5 Task 2.2: Propagation Delay Measurement of the Lamp Logic Circuit

After the setup in Task 2.1, the focus of this task was to measure the delay introduced by the full NAND gate implementation of the Lamp logic. The time between the rising edge of the input square wave and the corresponding output edge was recorded. Any minor overshoot or ringing on the output signal was disregarded, as the delay measurement was taken strictly from the point of logic-level threshold crossing on both input and output.

Measurement Method

- The square wave applied through the signal generator simulated repeated switch presses.
- Delay was observed using the time difference between the signals on CH1 (input) and CH2 (output).
- Both traces were captured using a Tektronix oscilloscope.

Result

The measured propagation delay from input to output in the NAND-based Lamp circuit was:

16.400 ns

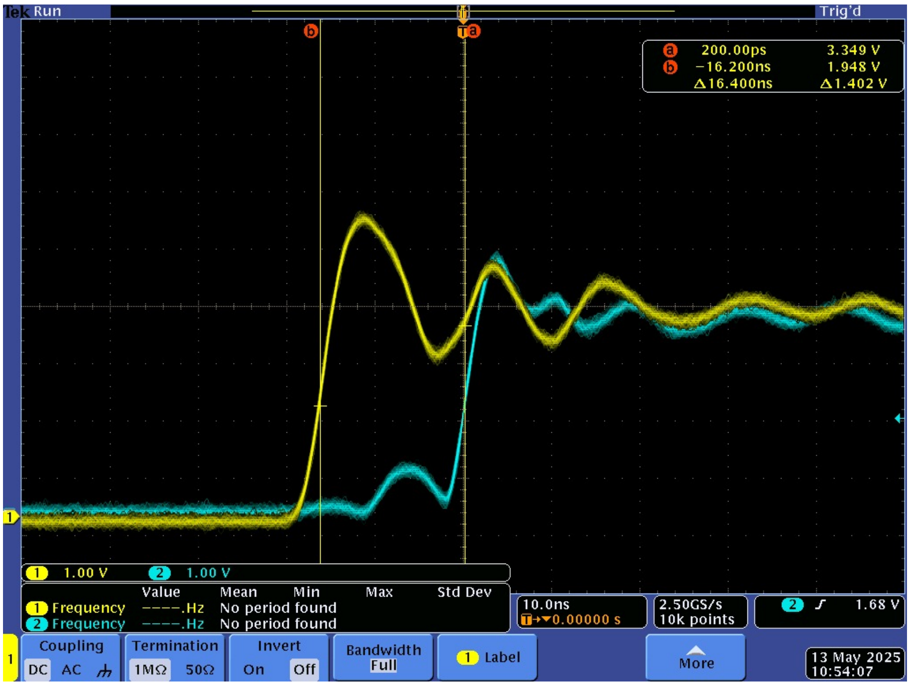

Figure 7: Oscilloscope screenshot showing input (CH1) and output (CH2) waveforms and delay measurement

Discussion

Although the complete Lamp logic circuit is composed of multiple interconnected NAND gates, the delay measurement in this task specifically focused on the signal path from the `SWITCH` input to the `LIGHT` output. Based on the circuit implementation, the `SWITCH` signal passes through exactly two NAND gates in the active logic branch: one to perform a NAND operation with `NOT SC2`, and another to invert the result (i.e., emulate AND behavior using NANDs). Therefore, the measured propagation delay of **16.4 ns** can be attributed to the combined effect of these two NAND gates. This matches typical expectations for 74xx series NAND gate delays and confirms the accuracy of the experimental setup.

6 Task 2.3: Propagation Delay Measurement of a Single NAND Gate

In this task, we disassembled the original Lamp control circuit and constructed a simple test circuit using a single 2-input NAND gate configured as a NOT gate. This allowed us to isolate the behavior of a single logic element and measure its propagation delay accurately.

Method

- The function generator provided a square wave with:
  - 3.3 V peak-to-peak
  - 1.65 V offset
  - 10 kHz frequency
- The square wave input was applied to both inputs of the NAND gate to form a NOT gate.
- The oscilloscope channels were connected as follows:
  - CH1: Function generator input
  - CH2: NAND gate output

Measurement Result

The propagation delay between the input and output signals was measured on the oscilloscope. The delay corresponds to the time it takes for the NAND gate to respond to a change in input.

Measured Propagation Delay: 9.12 ns

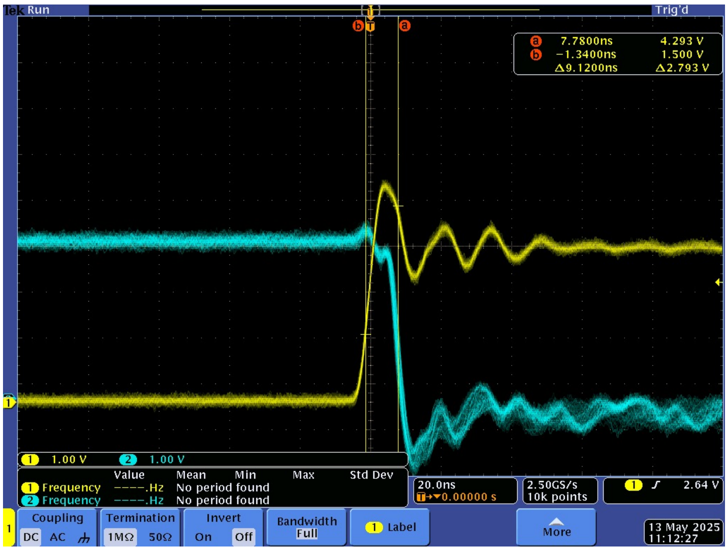

Figure 8: Oscilloscope screenshot showing delay through a single NAND gate used as inverter

Conclusion

Compared to the delay measured in Task 2.2 for the Lamp logic circuit (16.4 ns), the delay through a single NAND gate is significantly lower. This validates that the observed delay in the larger circuit was caused by the cumulative effect of multiple NAND gates (in that case, 2 gates for the path from `SWITCH` to `LIGHT`). The result from Task 2.3 also aligns well with typical propagation delays for 74xx series CMOS NAND gates in inverter configuration.

7 Task 3.1: Vivado Settings and Adjustments

- A new RTL project was created in Vivado.
- The Room Controller VHDL file (design only, no testbench) was added to the project.
- The IOM board switches and LEDs were mapped to the corresponding VHDL ports for functional testing.
- The XDC constraints file from Moodle was added to the project.
- Signal names in the XDC file were updated to match the entity port names in the VHDL file.
- Only the switches and LEDs used in the design were enabled in the XDC file; all others were commented out using the # symbol.
- Pin numbers were left unchanged as instructed.

8 Task 3.2: Simulated Schemes by Vivado

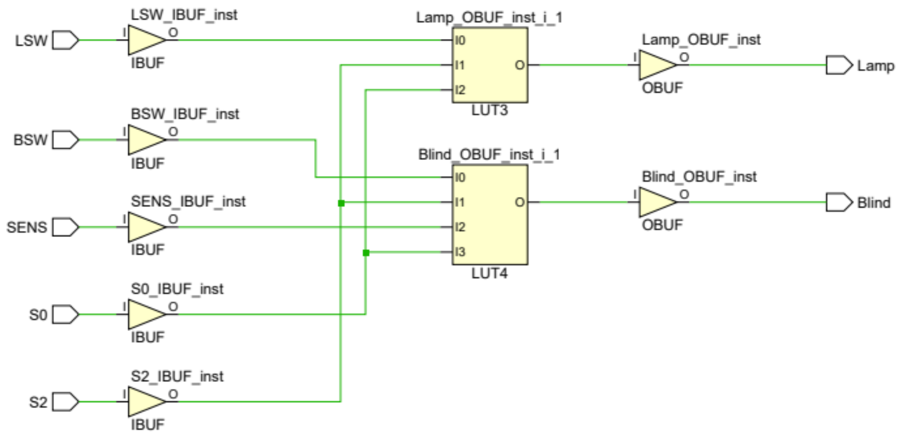

Figure 9: Technology schematic view after synthesis

- This view shows how the VHDL logic was synthesized into physical FPGA resources.
- Inputs are connected through IBUF primitives, and outputs through OBUFs.
- Vivado optimized the logic expressions efficiently into compact LUTs.

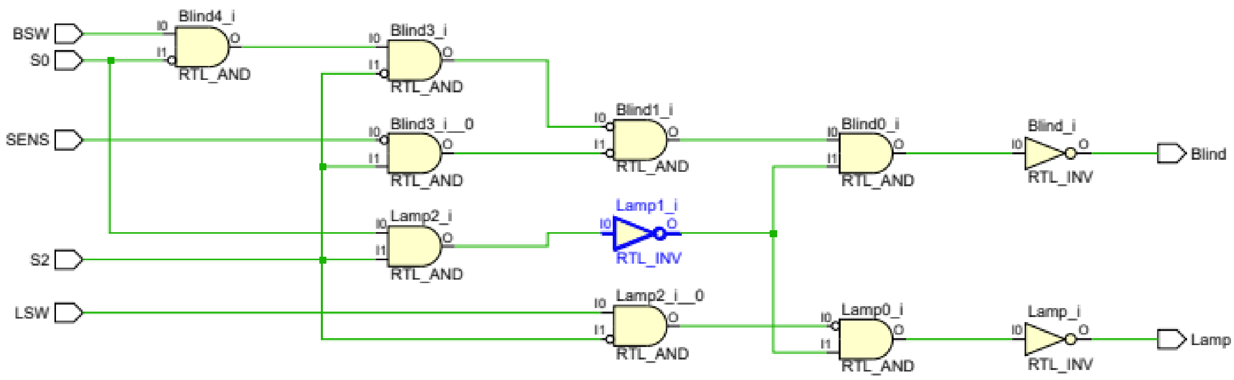

Figure 10: RTL schematic view of logical structure

- This RTL diagram shows the structural interpretation of the original VHDL code.
- It includes gates like ANDs and NOTs used to build the Lamp and Blind logic branches.
- The circuit follows the expected logic behavior and matches the design based on our truth table.

9 Task 3.3: Performed Test Plan in the FPGA Board

The generated bitstream was successfully programmed onto the FPGA board. We tested all input combinations using the IOM board switches and observed the corresponding LED outputs. Each of the predefined test cases (R1.1 to R4.4) was verified according to the truth table.

10 Resources and Literature

References

[1] P. Schulz, *Microsoft PowerPoint - DI_S25_English_v_0.9.pptx*, Department of Information and Electrical Engineering, Faculty of Engineering and Computer Science, HAW Hamburg, Hamburg, 2024.

[2] P. Schulz, *VHDL_EDAplayground_Tutorial.English.pdf*, Department of Information and Electrical Engineering, Faculty of Engineering and Computer Science, HAW Hamburg, Hamburg, 2025.

[3] EDAPlayground, *EDA Playground Online VHDL/Verilog Simulator*, accessed May 2025. Available: `https://www.edaplayground.com`

[4] Xilinx, *Vivado Design Suite User Guide: Tutorial (UG888)*, accessed May 2025. Available: `https://www.xilinx.com/support/documentation-navigation/design-hubs/dh0002-vivado-design-hub.html`

[5] Xilinx, *Texas Instruments 7400 Quad 2-Input NAND Gate*, accessed May 2025. Available: `https://www.ti.com/lit/ds/symlink/sn74ls00.pdf`

[6] Digital, *Digital Logic Simulation Software (digital.exe)*, Version X.X, 2025. Available: `https://github.com/hneemann/Digital`